0. **Code:** Kusumanjali(Final test).zip

1. FINAL TEST / ConsoleApp1 / ConsoleApp1 / Program.cs
2. FINAL TEST / ConsoleApp2 / Program.cs
3. FINAL TEST / ConsoleApp3 / Program.cs

1. FINAL TEST / ConsoleApp1 / ConsoleApp1 / Program.cs

```
namespace ConsoleApp1
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Enter an integer:");
            int n = Convert.ToInt32(Console.ReadLine());
            Console.WriteLine("Enter the base-N");
            int m = Convert.ToInt32(Console.ReadLine());
            int remainder = 0;
            int sum = 0;
            int largestdigit = int.MinValue;

            while (true)
            {
                if (m ≤ 0)
                {
                    Console.WriteLine("Enter the base-N");
                    m = Convert.ToInt32(Console.ReadLine());
                }
                if (m ≥ 1)
                {
                    remainder = n % m;
                    sum = sum + remainder;
                    if (remainder > largestdigit)
                    {
                        largestdigit = remainder;
                    }

                }
```

A1

A5 When n=0, largestDigit is minValue? It should be 0.

(n>0) A3

If (n<0) n=-n;

A3

OK

Continue

A2

A4

A5

n=n/m; A3

1. **FINAL TEST / ConsoleApp1 / ConsoleApp1 / Program.cs**

```
            }

            Console.WriteLine("Sum of the base-N digits" + sum);
            Console.WriteLine("Maximum digit is:" + largestdigit);

        }


    }
}
```

2. FINAL TEST / ConsoleApp2 / Program.cs

```
namespace ConsoleApp2
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] array = {
                23,
                1,
                10,
                5,
                2
            };

            int[] result = Method(array);
            foreach(int count in result)
            {
                Console.WriteLine(count);
            }

        }
        public static int[] Method(int[] array)
        {
            int n = array.Length;
            int[] result = new int[n];
            for (int i = 0; i < n; i++)
            {
                int key = array[i];
                //compare each element with one before it
                int j = i - 1;
```

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132

2. **FINAL TEST / ConsoleApp2 / Program.cs**

```
                //Move an element greater than the key element one position ahead
                while (j ≥ 0 && array[j + 1] ≥ key)
                {
                    result[i]++;
                    j--;
                }
            }
            return result;
        }
    }
}
```

if(result[j+1] ≥ key) {

}

No

When you are not swapping elements, you need to compare from the beginning.

B3

3. FINAL TEST / ConsoleApp3 / Program.cs

```
namespace ConsoleApp3
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] array = {
                1,
                2,
                3,
                2,
                7,
                3,
                4,
                4,
                5,
                5,
                4,
                5,
                1,
                6,
                1,
                1
            };

        }
        public static int[] Method(int[] array)
        {
            if (array == null)
            {
```

Incompl

int

3. **FINAL TEST / ConsoleApp3 / Program.cs**

```
            return null;
        }
        //frequency count
        Dictionary < int, int > dict = new Dictionary < int, int >
          ();
        //largest numbers
        Dictionary < int, int > dict1 = new Dictionary < int, int >
          ();

        foreach(int num in array)
        {
            if (dict.ContainsKey(num))
            {
                dict[num]++;
            }
            else
            {
                dict[num] = 1;
            }
            if (!dict1.ContainsKey(dict[num]) || dict1[dict[num]] < num)
            {
                dict1[dict[num]] = num;
            }
        }
        int sum = 0;
        foreach(var kvp in dict1)
        {
            sum += kvp.Key * kvp.Value;
        }
        return sum;
```

C2

C3

C4 ok

(Separate loop — You can only get the max ace you have a count dict

3. FINAL TEST / ConsoleApp3 / Program.cs

```
61
62          }
63
64
65      }
66  }
```